

**SAFETY DATA SHEET**

Creation Date 11-July-2014

Revision Date 28-March-2024

Revision Number 4

**1. Identification**

**Product Name** Mercury(II) oxide, Puratronic®

**Cat No. :** 10810

**CAS-No** 21908-53-2  
**Synonyms** Mercuric oxide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |            |
|-------------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                                  | Category 2 |
| <b>Acute dermal toxicity</b>                                | Category 1 |
| <b>Acute Inhalation Toxicity</b>                            | Category 2 |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2 |
| Target Organs - Urinary Tract.                              |            |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Fatal if swallowed, in contact with skin or if inhaled

May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Do not get in eyes, on skin, or on clothing  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Take off immediately all contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects  
Light sensitive

### 3. Composition/Information on Ingredients

| Component           | CAS-No     | Weight % |
|---------------------|------------|----------|
| Mercury oxide (HgO) | 21908-53-2 | 100      |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.                                                                                                                                                                                 |
| <b>Inhalation</b>                      | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Most important symptoms/effects</b> | No information available.                                                                                                                                                                                                                                                                                                      |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Very toxic. Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Oxygen.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
4

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Wear self-contained breathing apparatus and protective suit. Evacuate personnel to safe areas. Ensure adequate ventilation. Avoid dust formation. Do not get in eyes, on skin, or on clothing.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up** Wear self-contained breathing apparatus and protective suit. Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Use only under a chemical fume hood. Wear personal protective equipment/face protection. Avoid dust formation. Do not breathe dust. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Chlorine. Peroxides. Metals. Reducing Agent. Finely powdered metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component           | Alberta                              | British Columbia                     | Ontario TWAEV                        | Quebec                               | ACGIH TLV                            | OSHA PEL                                    | NIOSH                                                                                       |
|---------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| Mercury oxide (HgO) | TWA: 0.025 mg/m <sup>3</sup><br>Skin | TWA: 0.025 mg/m <sup>3</sup><br>Skin | TWA: 0.025 mg/m <sup>3</sup><br>Skin | TWA: 0.025 mg/m <sup>3</sup><br>Skin | TWA: 0.025 mg/m <sup>3</sup><br>Skin | (Vacated)<br>Ceiling: 0.1 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup><br>Ceiling: 0.1 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Physical State</b>                   | Powder Solid             |
| <b>Appearance</b>                       | Yellow-orange            |
| <b>Odor</b>                             | Odorless                 |
| <b>Odor Threshold</b>                   | No information available |
| <b>pH</b>                               | No information available |
| <b>Melting Point/Range</b>              | 500 °C / 932 °F          |
| <b>Boiling Point/Range</b>              | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Evaporation Rate</b>                 | Not applicable           |
| <b>Flammability (solid,gas)</b>         | No information available |
| <b>Flammability or explosive limits</b> |                          |
| Upper                                   | No data available        |

|                                        |                          |
|----------------------------------------|--------------------------|
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Hg O                     |
| Molecular Weight                       | 216.59                   |

## 10. Stability and reactivity

|                                  |                                                                     |
|----------------------------------|---------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                          |
| Stability                        | Stable under recommended storage conditions. Light sensitive.       |
| Conditions to Avoid              | Avoid dust formation. Exposure to light. Incompatible products.     |
| Incompatible Materials           | Chlorine, Peroxides, Metals, Reducing Agent, Finely powdered metals |
| Hazardous Decomposition Products | Oxygen                                                              |
| Hazardous Polymerization         | No information available.                                           |
| Hazardous Reactions              | None under normal processing.                                       |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component           | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|---------------------|-------------------------|-------------|-----------------|
| Mercury oxide (HgO) | LD50 = 18 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component           | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------|------------|------------|------------|------------|------------|------------|
| Mercury oxide (HgO) | 21908-53-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known  
**STOT - repeated exposure** Urinary Tract

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1641  
 Proper Shipping Name MERCURY OXIDE  
 Hazard Class 6.1  
 Packing Group II

### TDG

UN-No UN1641  
 Proper Shipping Name MERCURY OXIDE  
 Hazard Class 6.1  
 Packing Group II

### IATA

UN-No UN1641  
 Proper Shipping Name MERCURY OXIDE  
 Hazard Class 6.1  
 Packing Group II

### IMDG/IMO

UN-No UN1641  
 Proper Shipping Name MERCURY OXIDE  
 Hazard Class 6.1  
 Packing Group II

## 15. Regulatory information

### International Inventories

| Component           | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Mercury oxide (HgO) | 21908-53-2 | X   | -    | X    | ACTIVE                                        | 244-654-7 | -      | -   |

| Component           | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Mercury oxide (HgO) | 21908-53-2 | X     | KE-23130 | X    | X    | X    | X    | X     | X     |

Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component           | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA)<br>- List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|---------------------|------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------|
| Mercury oxide (HgO) | Part 1, Group B Substance                            |                                                                               |                                           |

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component           | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Mercury oxide (HgO) | -                                                                   | Use restricted. See item 18.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component           | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Mercury oxide (HgO) | 21908-53-2 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component           | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Mercury oxide (HgO) | 21908-53-2 | Not applicable                                                                            | Not applicable                                                                           | X                          | Annex I - Y29                      |

**16. Other information****Prepared By**

Product Safety Department  
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[www.thermofisher.com](http://www.thermofisher.com)

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**Print Date**

28-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**